

Oil Price Shocks and Inflation Dynamics

A Structural VAR (SVAR) Approach

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The Research Question

To what extent do oil price shocks contribute to inflation dynamics, and how much does removing these shocks change observed inflation?

Oil Shock Proxy

- WTI crude oil price changes
- FRED series: OILPRICEx
- Well-established proxy in the monetary policy literature (Bernanke, Gertler & Watson 1997)

Why It Matters

- Central banks must separate transitory supply shocks from persistent demand pressure
- Misidentifying shocks leads to over-reaction
 - raising rates when not needed

Supply vs Demand Shocks Require Different Policy Responses

SUPPLY SHOCK

- Cost-push: same output, higher prices
- Short-run supply curve shifts upward
- Effect is transitory - self-correcting
- Policy lesson: look through it

Traditional core: excludes oil & food categories regardless of the underlying shock - economically arbitrary

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa x_t + u_t^s$$

DEMAND SHOCK

- Demand-pull: higher output and higher prices
- Demand curve shifts outward
- Effect is persistent - does not self-correct
- Policy lesson: act decisively

SVAR core (Quah & Vahey 1995): removes only the component with no long-run output effect - economically grounded

Building on Established Literature

1980

Sims

Foundational VAR methodology for macroeconomic systems

1995

Quah & Vahey

Pioneered SVAR-based core inflation - the component with no long-run output effect

1997

Bernanke, Gertler & Watson

Established oil price changes as a valid supply shock proxy

2002

Stock & Watson

Macroeconomic forecasting using diffusion indexes

2007

Blanchard & Galí

Documented the declining real effects of oil shocks in modern economies

Our Contribution: 4-variable recursive SVAR on FRED-MD (Feb 1959 – Feb 2026) · Oil as structural instrument · Full sample comparison with headline inflation

Data & Model Specification

Monthly FRED-MD data, Feb 1959 – Feb 2026

Variable	FRED Code	Role	Transform
Oil Price (WTI)	OILPRICE	Supply shock	$\Delta \log$
Industrial Production	INDPRO	Output proxy	$\Delta \log$
CPI	CPIAUCSL	Inflation	$\Delta \log \times 1200$
Fed Funds Rate	FEDFUNDS	Policy rate	Levels

VAR(2) Model

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + c + \varepsilon_t$$

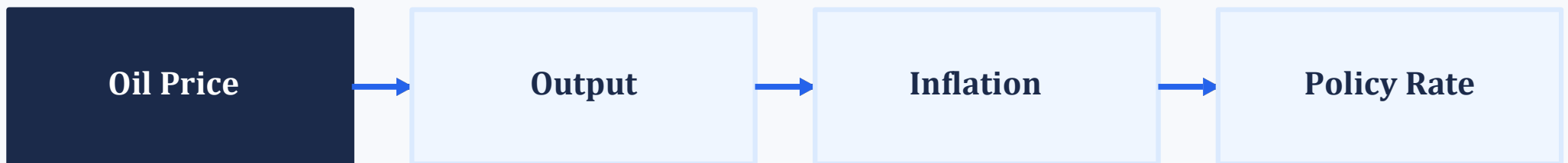
$Y_t = [\text{oil}, \text{output}, \text{inflation}, \text{rate}]^\top$

Lag length: 2 (BIC & HQ criteria)

Stability Check

- All characteristic roots inside unit circle
- Log-differences ensure stationarity
- Interest rate in levels (policy instrument)
- Final clean dataset: **803 obs.**

Identification & Three-Step Analysis



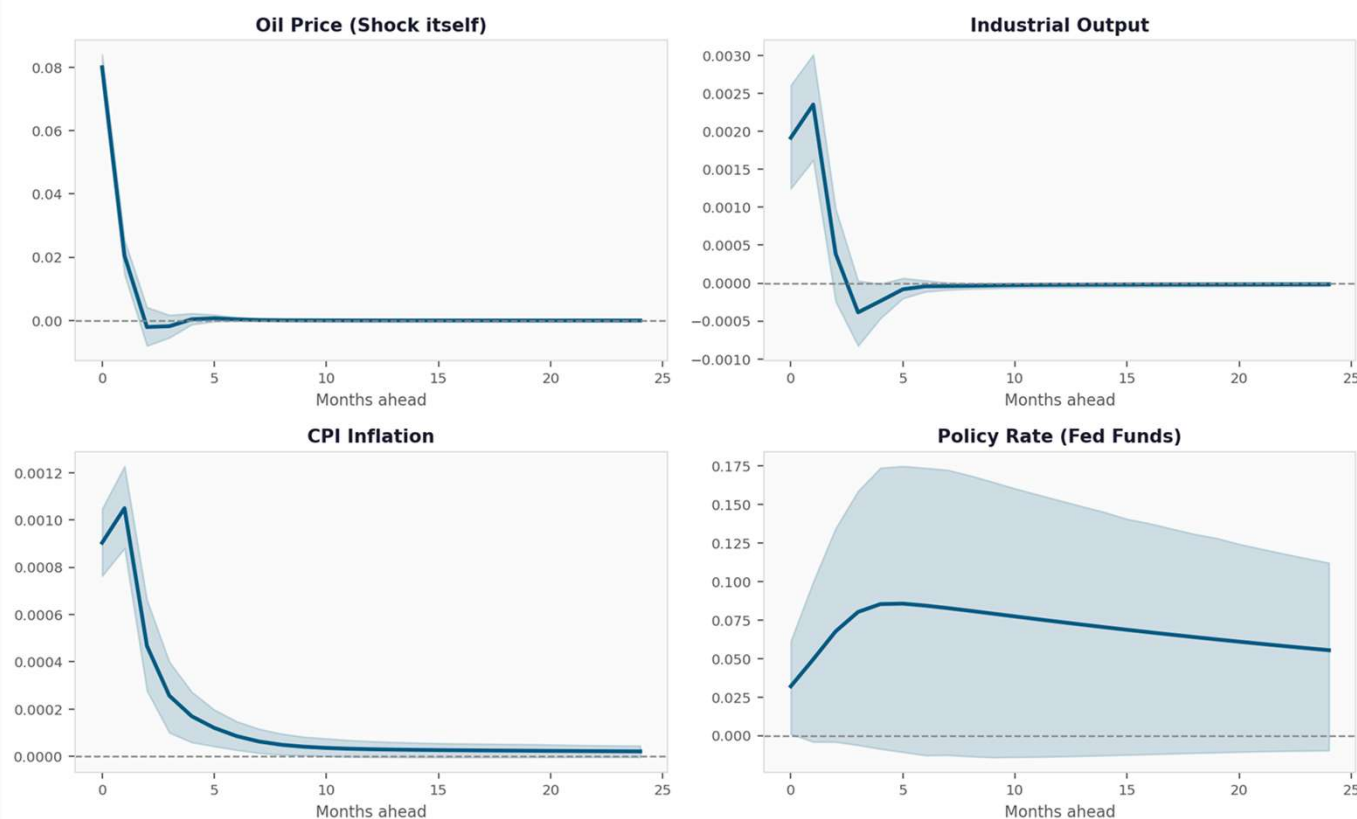
Cholesky Ordering - oil is contemporaneously exogenous to all domestic variables

- 01 Impulse Response Functions (IRF)**
Trace inflation, output & policy rate response to a 1-SD oil shock over 25 months. Determines whether the shock is transitory or persistent.
- 02 Forecast Error Variance Decomposition (FEVD)**
What fraction of inflation's variance comes from each structural shock at horizons 1–24 months? Quantifies relative importance.
- 03 Oil-Adjusted Inflation**
Oil-adjusted inflation = Headline - estimated oil shock contribution (convolution with IRF). Compared to headline over the full 1960–2026 sample.

Oil Shocks Have No Persistent Inflation Effect

IRF · 1-SD oil shock · 25-month horizon · 95% confidence interval

Impulse Responses to a One-Standard-Deviation Oil (Supply) Shock



Key Findings

⚡ Oil shock decays within 3 - 4 months → transitory

≈ 0 Output response near-zero - minimal real effects

✗ Inflation response statistically insignificant (CI crosses zero immediately)

↑ Fed reacts to oil shocks persistently even without inflationary pass-through

→ Consistent with Blanchard & Galí (2007)

Inflation Is Mostly Self-Driven (~60–84%)

~25%

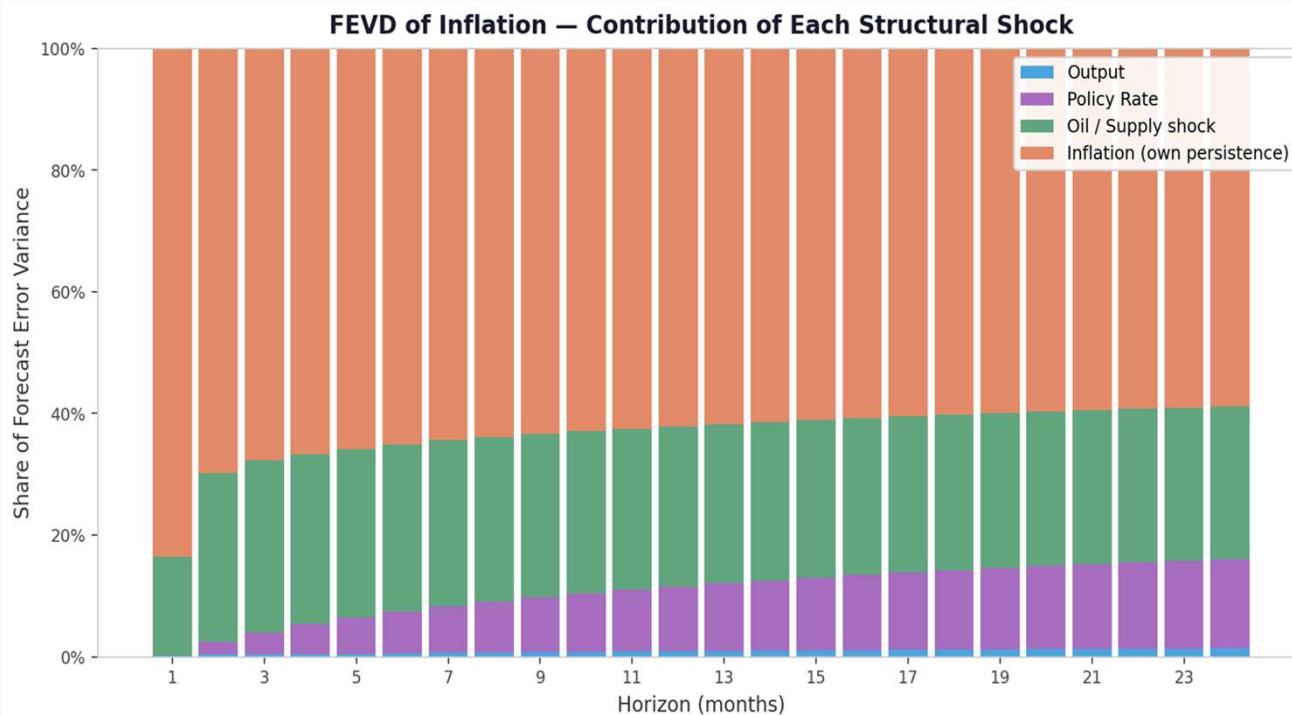
explained by oil shocks
(medium horizon)

~84%

own-persistence at
1-month horizon

<1.5%

output shocks
(negligible)



Horizon	Oil	Inflation	Policy
1m	16.2%	83.6%	0.0%
3m	28.3%	67.7%	3.5%
6m	27.5%	65.1%	6.9%
12m	26.3%	62.1%	10.6%
24m	25.0%	58.9%	14.8%

Building an Oil-Adjusted Inflation Measure

A four-step pipeline

$$\text{Oil-Adjusted Inflation}_t = \text{Headline Inflation}_t - \sum_h \psi_h \cdot \text{OilShock}_{t-h}$$

1

Estimate VAR(2)

4-variable system: oil, output, inflation, fed funds. Lag 2 by BIC/HQ.

2

Compute IRF

Extract orthogonal impulse response of inflation to oil shock: ψ_h for $h = 0, \dots, 12$

3

Convolution Filter

Dynamic oil contribution at each date: $\sum_h \psi_h \cdot \text{oil}_{t-h}$ over a 12-month window

4

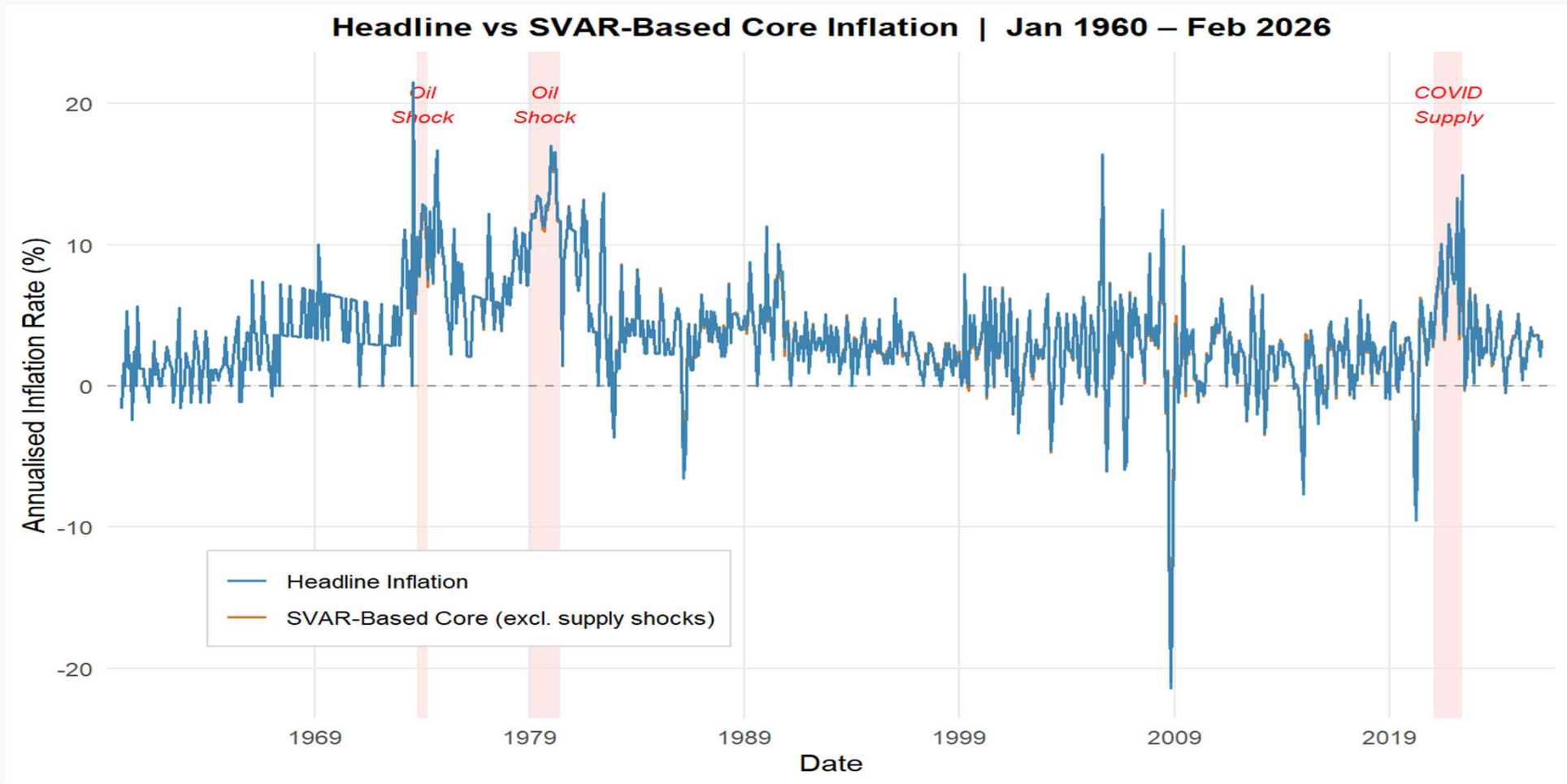
Subtract & Compare

Oil-adjusted = Inflation – oil contribution. Compare to headline (1960–2026).

All steps are data-driven: no arbitrary category exclusions. The only inputs are the structural SVAR and oil price history.

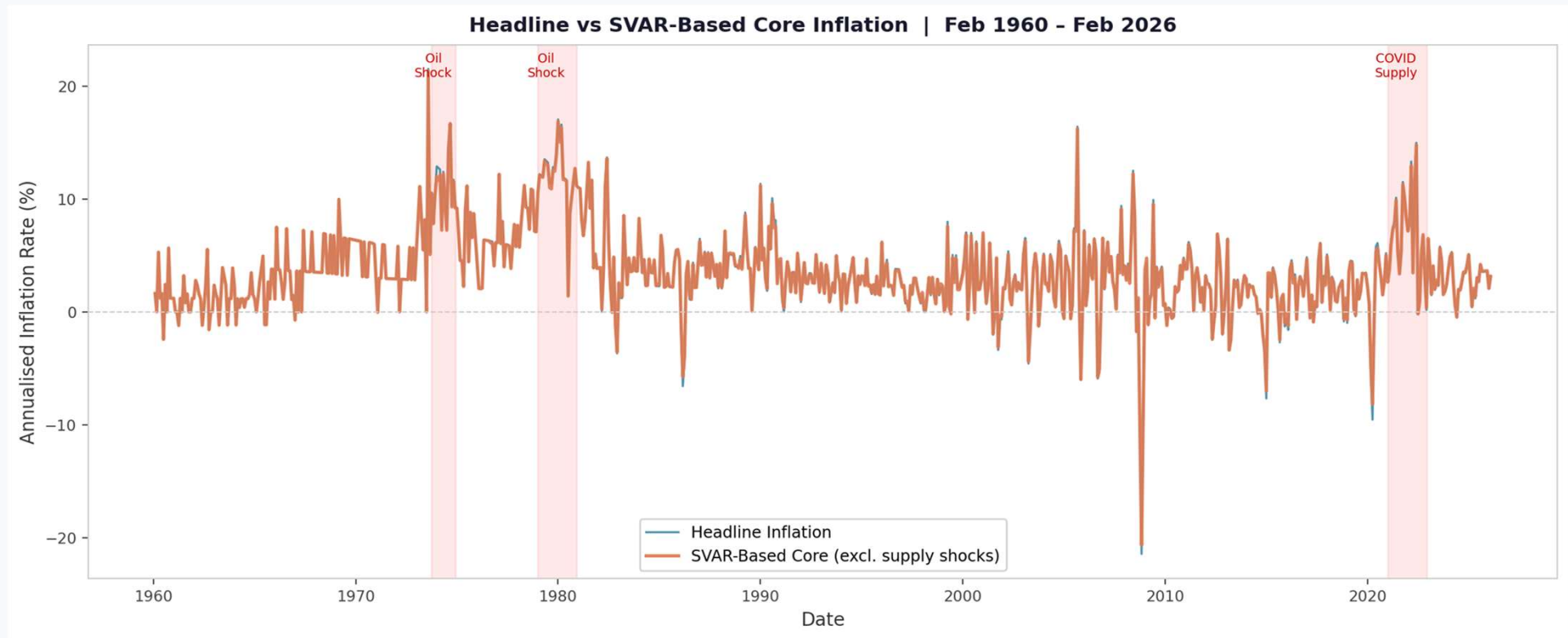
Oil-Adjusted Inflation Closely Tracks Headline

Feb 1960 - Feb 2026 · Shaded = major supply shock episodes



Oil-Adjusted Inflation Closely Tracks Headline

Feb 1960 - Feb 2026 · Shaded = major supply shock episodes



↓ Oil-adjusted slightly smoother - oil spikes filtered out

Same long run trend, removing shocks doesn't change underlying path

Synthesising the Evidence

Oil based supply shocks are transitory

- IRF: Oil shock dissipates within 3 - 4 months
- Direct inflationary effect indistinguishable from zero (Conf. Interval crosses zero)
- Consistent with Blanchard & Galí (2007) on declining oil sensitivity

Oil based supply shocks are not irrelevant

- FEVD oil explains:
~16% of inflation variance at 1 month,
~ 25-28% at 6 months
- Indirect effects and uncertainty persist even if direct transmission is weak

Inflation is mainly self-perpetuating

- Own-persistence explains 84% at horizon 1, 59% at 24 months
- This dominance implies sustained tightening is needed once inflation is entrenched
- Oil-adjusted measure confirms this - removing oil shocks leaves inflation path unchanged.

Three Policy Takeaways for Central Banks

Central banks should look through transitory oil shock inflation - focus on persistence, not level

1

Do Not Over-React to Oil Shocks

Oil-shock inflation is transitory. Rate hikes in response to supply-driven spikes risk strangling real activity without reducing underlying inflation pressure.

Evidence: IRF shows near-zero, insignificant inflation response

2

Focus on Inflation Persistence

~60–84% of inflation variance reflects its own dynamics. The key signal is whether inflation is becoming entrenched, not whether oil prices rose last month.

Evidence: FEVD - own-persistence dominates at all horizons

3

Use Oil-Adjusted Inflation as a Diagnostic Tool

Oil-adjusted inflation provides a theoretically grounded filter, removing the identified oil shock contribution. Useful as a cross-check alongside headline inflation.

Evidence: Oil-adjusted only marginally smoother → headline remains valid

Limitations & Future Extensions

Identification assumptions	Issue: Cholesky ordering assumes oil is contemporaneously exogenous. May not hold if domestic demand affects global oil prices	Fix: Sign restrictions (Uhlig 2005) or external instruments for more credible identification
Single supply proxy	Issue: Oil disruptions like the 2021 Suez Canal blockage not reflected in prices	Fix: Add NY Fed Global Supply Chain Pressure Index (GSCPI) or PPI for a richer supply block
Time-invariant parameters	Issue: Fixed-coefficient VAR spans Bretton Woods, Great Moderation, and ZLB which are very different regimes	Fix: Time-varying parameter VAR (TVP-VAR) to allow for structural breaks
Core construction method	Issue: Convolution uses only first-period oil shock contribution	Fix: Full historical decomposition (Kilian & Lütkepohl 2017) for precise structural attribution

The Big Picture

Oil-adjusted inflation is theoretically cleaner, but only marginally different in practice

Theoretically Superior

- Removes economically identified shocks, not arbitrary categories
- Grounded in Quah & Vahey (1995) structural framework
- Consistent with monetary policy objectives

Practically Modest

- Oil-adjusted series is only marginally smoother than headline
- 2 earlier panels show long-run trend that tell a similar story
- Policymakers using headline CPI are not badly misinformed

Limitations

- Single supply proxy (oil) misses other disruptions
- Time-invariant VAR spans very different monetary regimes
- Convolution method is less precise than full historical decomposition

Recommendation: deploy oil-adjusted inflation as a diagnostic cross-check alongside headline inflation, not as a replacement.

CONCLUSION

“Oil shocks matter - but do not dominate inflation dynamics.”

- 1** Oil shocks are transitory - IRF shows near-zero, statistically insignificant inflation response
- 2** Oil shocks explain ~25% of inflation variance, important as 2nd driver, but not dominant (FEVD)
- 3** Inflation is primarily driven by own-persistence (~60–84%): the main challenge for monetary policy
- 4** Oil-adjusted inflation is marginally smoother than headline - removes noise, not the underlying trend

Thank you
Any questions?
(references cited on next page)

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